SoftwareX 35 (2026) 102770

Contents lists available at ScienceDirect

SoftwareX

journal homepage: www.elsevier.com/locate/softx

Original Software Publication

App4WoSCCinR: a web-based bibliometric analysis platform for the web of science core collection

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ARTICLE INFO

Keywords:
Bibliometric software
WoSCC metadata
Visualization framework
AAC metric
Web-based application

ABSTRACT

Numerous bibliometric studies rely on specialized software to visualize Web of Science Core Collection (WoSCC) metadata, but direct, reproducible workflows for standard 72-column XLS files remain limited. We developed App4WoSCCinR, a web-based framework that enables users to upload WoSCC metadata and execute bibliometric analysis through a simple two-step process. Using SoftwareX articles published between 2017 and March 2026 as an illustrative dataset, the application generated a structured performance summary and visual outputs across countries, institutions, departments, authors, journals, keywords, and temporal trends. It also incorporates the Absolute Advantage Coefficient (AAC) to identify dominant entities within bibliometric networks. The demonstration showed that App4WoSCCinR can efficiently transform standard WoSCC metadata into interpretable summaries and visual analytics for research evaluation. Overall, the framework provides a practical and transparent solution for low-barrier bibliometric analysis directly from WoSCC spreadsheet exports.

- A web-based tool, App4WoSCCinR, was developed for streamlined bibliometric analysis from standard 72-column WoSCC metadata.
- The framework produced a summary report and over 100 visualizations across 10 metadata domains, incorporating AAC to quantify dominance.
- Analysis of 2,061 SoftwareX articles demonstrated the tool's practical value for transparent, efficient, and reproducible bibliometric workflows.

Metadata

Nr	Code metadata description	*Metadata*
C1	Current code version	*v1.0*
C2	Permanent link to code/repository used for this code version	https://github.com/smilechien/App4WoSCCinR/
C3	Permanent link to reproducible capsule	https://smilechien.shinyapps.io/woscc/
C4	Legal code license	*MIT License*
C5	Code versioning system used	*Git*
C6	Software code languages, tools and services used	*R*
C7	Compilation requirements, operating environments and dependencies	shiny, dplyr, tidyr, stringr, ggplot2, leaflet, readxl, DT, visNetwork, and bibliometrix.
C8	MP4 video	https://youtu.be/RbqA3KLAQjI
C9	Support email for questions	rasch.smile@gmail.com

1. Motivation and significance

Bibliometric analysis has become an important approach for evaluating research productivity, identifying emerging trends, mapping

Abbreviations: AAC, Absolute Advantage Coefficient; CSV, Comma-Separated Values; PNG, Portable Network Graphics; SSplot, Silhouette Plot; VBA, Visual Basic for Applications; WoS, Web of Science; WoSCC, Web of Science Core Collection; XLS, Excel Spreadsheet Format; XLSX, Office Open XML Spreadsheet Format; ZIP, Compressed Archive File.

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https://doi.org/10.1016/j.softx.2026.102770
Received 24 March 2026; Received in revised form 27 May 2026; Accepted 1 June 2026
Available online 4 June 2026
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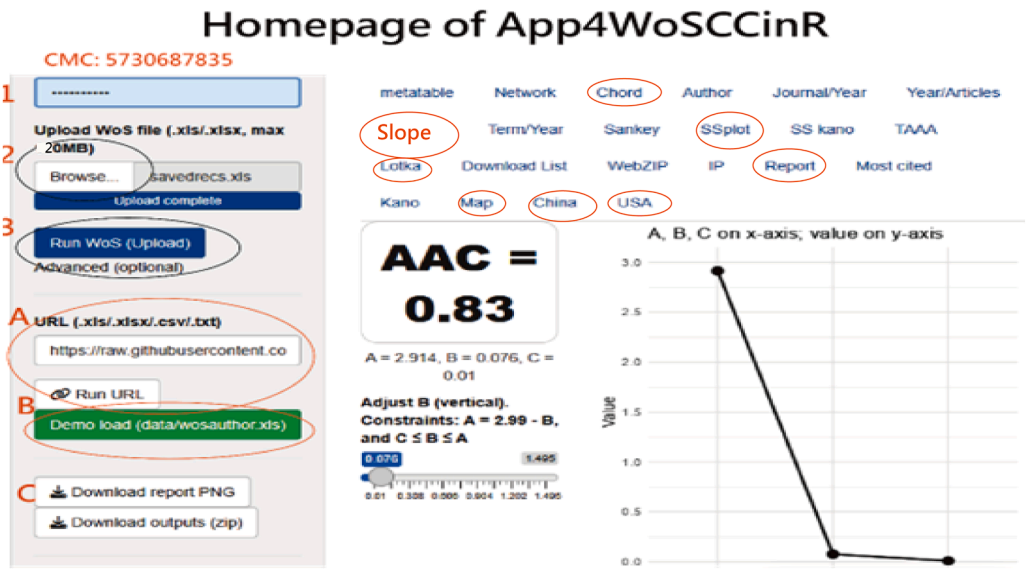


Fig. 1. Homepage of App4WoSCCinR illustrating the two-step workflow.

knowledge structures, and assessing scholarly impact across disciplines [1–3]. Widely used tools such as CiteSpace [4], VOSviewer [5], and Bibliometrix [6] enable visualization from standardized Web of Science Core Collection (WoSCC) data [7].However, these tools typically require specific input formats and predefined workflows, limiting flexibility when working with commonly available 72-column Excel-exported WoSCC files.

In practice, such 72-column files often require extensive manual preprocessing—including data cleaning, field standardization, and restructuring—before meaningful analysis can be conducted [8–12]. These steps are time-consuming, error-prone, and hinder reproducibility, particularly for users without programming expertise.

1.1. Existing tools and remaining workflow challenges

A mature software [4–6,13] ecosystem already supports such work with plain text files for non-coders. Reviews of bibliometric software have shown that these tools vary in scope, with different strengths in performance analysis, science mapping, preprocessing, visualization, and user accessibility [14].

Recent developments have further expanded this ecosystem. For example, BibexPy [15] was introduced as a preprocessing-oriented tool for merging WoSCC and Scopus data, DOI-based deduplication, and metadata enrichment, while BiblioMerge [16] was designed to support merging, duplicate removal, author normalization, and compatibility with downstream bibliometric programs. These developments show that important progress has already been made in automating bibliometric workflows, especially for data integration and cleaning.

Nonetheless, the availability of multiple tools does not eliminate a practical workflow challenge for many users(i.e., it still requires moving across several steps from exporting records and cleaning metadata to generating publication-ready Figures [17]. Even when robust tools are available, the overall process may remain fragmented for users who prefer a direct upload-and-run workflow on commonly downloaded spreadsheet files rather than multi-stage scripting or tool chaining. A practical need therefore remains for an integrated, accessible workflow that enables bibliometric visualization directly from WoSCC Excel metadata.

1.2. Practical need for direct and reproducible WoSCC workflows

WoSCC [7] remains one of the most widely used sources for bibliometric research because it indexes journals, conference proceedings, and books across a broad range of disciplines [15,16]. WoSCC records can be exported to spreadsheet formats, which are convenient for users but may still require cleaning, normalization, restructuring, or format conversion before consistent downstream analysis. Related guidance and workflow papers likewise emphasize the importance of combining and cleaning bibliometric data before visualization and interpretation [8].

This issue is not simply technical; it also affects accessibility and reproducibility. Code-based environments offer high flexibility, but users without programming experience may find preprocessing and multi-tool orchestration burdensome. Conversely, graphical tools may reduce the coding burden but still differ in import expectations, field handling, export options, or output types. Taken together, these workflow-level constraints suggest the value of a low-barrier framework that can accept a standard WoSCC export, perform core preprocessing steps transparently, and generate repro ducible bibliometric outputs within a single environment.

Additionally, reproducibility remains a major concern in bibliometric research. Many studies lack detailed reporting of data sources, including WoSCC subsets, coverage periods, and search dates, limiting replicability [18]. Furthermore, inconsistencies across databases—such as incomplete author–affiliation links, misclassified document types, and uneven identifier coverage—can introduce bias [19–24] on following visualizations.

1.3. Study goal and scope

To address this practical need, we developed App4WoSCCinR [25], a browser-based framework designed for users who want to move directly from a WoSCC spreadsheet export to automated bibliometric summaries and visual outputs with minimal manual intervention. In the current implementation, the platform is organized around a simple two-step workflow—data upload and automated execution—and produces both a structured summary report and a large set of exportable visualizations across multiple metadata domains.

Using articles indexed in WoSCC for *SoftwareX* as an illustrative case, the present study demonstrates how App4WoSCCinR supports metadata preprocessing, performance analysis, and Figure generation within a unified web workflow. The emphasis is therefore on implementation, usability, and reproducible output generation, while positioning the software as a practical complement to the existing bibliometric tool

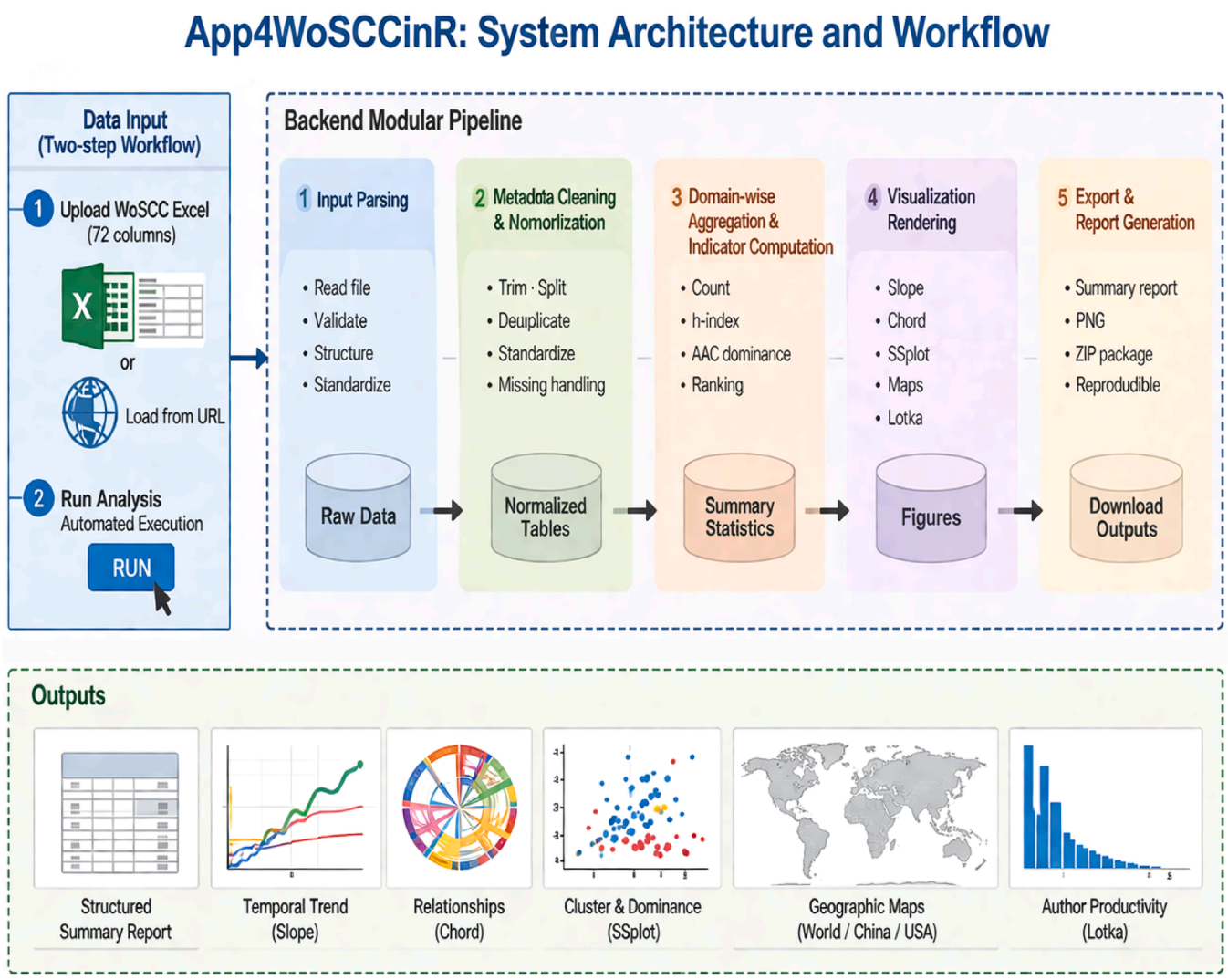


Fig. 2. System architecture and workflow of App4WoSCCinR, showing the transformation of WoSCC spreadsheet input into normalized intermediate tables, domain-level summary statistics, visualization outputs, and exportable reports.

ecosystem.

1.4. Contribution of the present work

The contribution of App4WoSCCinR lies in **workflow integration and accessibility**. Specifically, the framework emphasizes: (1) direct handling of WoSCC spreadsheet exports; (2) minimal manual preprocessing; (3) browser-based execution; (4) automatic generation of structured summary panels and diverse visualization outputs in a single run; (5) and AAC-enhanced dominance [26,27] reporting as a complementary layer beyond simple counts and rankings.

In this sense, the novelty of the software is practical rather than exclusive: it packages several otherwise separate steps into a streamlined web interface intended for transparent and reproducible bibliometric analysis.

2. Software description

App4WoSCCinR [25] is a web-based bibliometric analysis framework developed to process standard 72-column WoSCC metadata and generate both structured summary reports and multi-domain visualization outputs within a unified workflow.

The software is designed to reduce manual preprocessing and to support reproducible bibliometric analysis through a streamlined upload-and-run interface. Beyond the user-facing simplicity of the platform, the system is organized as a modular analytical pipeline that separates data ingestion, metadata normalization, indicator computation, visualization rendering, and export.

2.1. System architecture and workflow

The architecture of App4WoSCCinR consists of five major modules:

(1) **input parsing,**
(2) **metadata cleaning and normalization,**
(3) **domain-wise aggregation and indicator computation,**
(4) **visualization rendering,** and
(5) **export and report generation.**

The overall workflow begins with the upload of a WoSCC spreadsheet file (.xls or .xlsx) or the provision of a valid URL pointing to bibliometric data.

The imported records are parsed into an internal tabular structure and then passed to preprocessing routines that clean, standardize, and restructure key metadata fields. The normalized outputs are subsequently converted into domain-specific summary tables, which are used to compute bibliometric indicators and generate the final visualizations and summary report. The resulting outputs can then be displayed in the

web interface or exported as image and archive files.

The two-step workflow (Fig. 1) —data upload and automated execution—serves as the entry point to a deeper modular backend that transforms raw WoSCC spreadsheet metadata into publication-ready Figures.

2.2. Modular design

Five modules are involved in App4WoSCCinR, as shown in Fig. 2.

2.2.1. Module 1: Input parsing

The input module accepts WoSCC metadata in spreadsheet form and supports both local file upload and URL-based loading. A demonstration dataset is also provided for rapid testing and illustration. Regardless of input source, all datasets are processed through the same parsing logic so that subsequent analytical steps remain consistent.

The software validates the file format, reads the raw worksheet content, checks that the imported structure is compatible with WoSCC-style metadata, and converts the records into a standard internal data frame for downstream processing.

2.2.2. Module 2: Metadata cleaning and normalization

The second module performs metadata preprocessing, including trimming, delimiter harmonization, missing-value handling, and restructuring of multi-value fields such as authors, affiliations, and keywords. This step is necessary because WoSCC spreadsheet exports, although convenient for users, often require normalization before consistent bibliometric analysis can be performed.

The preprocessing routines generate structured intermediate tables suitable for domain-level aggregation. For example, author, institution, department, country, and keyword fields are split, standardized, and transformed into countable analysis units. This module therefore acts as the bridge between raw WoSCC metadata and analysis-ready bibliometric objects.

2.2.3. Module 3: Domain-wise aggregation and indicator computation

After normalization, the third module constructs summary tables across major metadata domains, including 10 metadata doamins. For each domain, the software computes bibliometric indicators such as publication count and h-index, and applies the Absolute Advantage Coefficient (AAC) [26,27] to quantify the relative dominance of leading entities.

By separating aggregation from preprocessing, the software allows multiple visual outputs to be generated from the same standardized intermediate data rather than repeatedly transforming the original spreadsheet. This design improves consistency, efficiency, and traceability across outputs.

2.2.4. Module 4: Visualization rendering

The rendering module converts domain-level summary tables into graphical outputs. In the current implementation, App4WoSCCinR produces more than 100 visualizations spanning multiple metadata domains. These include summary plots[8], slope graphs, chord diagrams [28], Sankey-style relationship displays[29], Slope graphs[30], SSplots [31], world maps, regional maps for China and the United States, and Lotka-related author productivity displays.

Different metadata structures are routed to different rendering functions. Temporal summaries are used to generate slope plots[30]; relational or co-occurrence data are used in chord and network-based displays; geographic tables are passed to map functions; and clustered data are used to construct SSplots[31]. This modular rendering approach allows the platform to support heterogeneous visualizations within a single analytical environment.

2.2.5. Module 5: Export and report generation

The final module assembles the outputs into downloadable artifacts. A structured summary report is generated to present key contributors and bibliometric indicators across metadata domains, while individual Figs. can be exported as PNG files. In addition, a ZIP package of outputs can be downloaded to support reuse, sharing, and reproducibility.

This module ensures that App4WoSCCinR functions not only as an exploratory interface but also as a reporting tool capable of producing reusable materials for bibliometric research and presentation.

2.3. Implementation layers

From an implementation perspective, App4WoSCCinR can be viewed as four interacting layers:

(1) a **front-end Shiny interface,**
(2) a **server-side preprocessing layer,**
(3) an **analytical engine, and**
(4) a **Figure/Report export layer.**

The front-end interface manages file upload, URL input, execution, preview, and downloading. The server-side preprocessing layer handles raw-data parsing and metadata normalization. The analytical engine computes domain-level summaries and bibliometric indicators, including AAC-based dominance [26,27] reporting. The rendering and export layer then transforms these intermediate results into Figures and downloadable reports.

This layered design clarifies that the two-step user workflow is intentionally simple, whereas the software itself implements a structured backend pipeline for bibliometric data transformation and output generation.

2.4. Representative code logic

To improve readability and reproducibility, the representative implementation is now presented as formatted R code listings. The native R pipe operator |> passes the output of the expression on its left to the function on its right, allowing sequential data-processing steps to be written clearly.

2.4.1. WoSCC field parsing

The parser imports the uploaded WoSCC spreadsheet, standardizes column names, and trims character fields before downstream processing.

```
raw <- readxl::read_excel(input$file$datapath)
raw <- raw |>
  dplyr::rename_with(trimws) |>
  dplyr::mutate(
    dplyr::across(
      where(is.character),
      trimws
    )
  )
```

2.4.2. Author and affiliation normalization

Multi-value metadata fields are separated into atomic units so that authors, affiliations, keywords, and related WoSCC fields can be counted consistently.

```
author_tbl <- raw |>
  dplyr::select(AU) |>
  tidyr::separate_rows(AU, sep = ";") |>
  dplyr::mutate(
    AU = stringr::str_squish(AU)
  ) |>
  dplyr::filter(AU != "")
```

The same normalization strategy is applied to affiliations, keywords,

Table 1
Summary report of leading entities across WoSCC metadata domains.

Domain	Top entities (Top 5)	h-index	Publications (n)	AAC
Country	United States; Spain; Germany; Poland; Italy	30; 13; 22; 13; 17	304; 193; 167; 156; 144	0.58
Institute	Delft University of Technology; National Institute of Telecommunications; University of Seville; AGH University of Krakow; Idaho National Laboratory	6; 5; 2; 4; 6	16; 14; 14; 13; 13	0.53
Department	Computer Science; Mechanical Engineering; Mathematics; Electrical and Computer Engineering; Physics	11; 7; 5; 7; 8	89; 26; 25; 22; 20	0.77
Author	Top 5 authors	6; 3; 4; 3; 2	16; 6; 6; 6; 5	0.73
Journal	SoftwareX	54	2061	0
Year	2025; 2024; 2023; 2022; 2020	7; 10; 17; 23; 30	474; 358; 316; 214; 165	0.54
Document type	Article; Correction; Software Review; Review; Editorial Material	54; 0; 3; 3; 1	2020; 18; 16; 6; 1	0.99
Discipline	Computer Science	54	2061	0.5
Keyword	MODEL; ALGORITHM; DESIGN; SIMULATION	13; 7; 14; 10	105; 38; 35; 26	0.72
Region	Texas; Florida; Idaho; Michigan; Beijing	6; 6; 6; 5; 2	15; 13; 13; 13; 11	0.54

Note. AAC = Absolute Advantage Coefficient. h-index and publication counts are shown for the leading entities within each metadata domain. AAC values closer to 1 indicate stronger dominance of the leading entity relative to nearby competitors.

countries, and other multi-value metadata fields. This ensures that summary tables are derived from standardized analytical units rather than raw spreadsheet strings.

2.4.3. AAC-based dominance computation

After domain-level counts are generated, App4WoSCCinR appends summary indicators, including the h-index [32] and AAC [8,26,27,33], to identify dominant entities within each bibliometric domain. The following listing illustrates author-level summarization.

```
summary_tbl <- author_tbl |>
  dplyr::count(AU, sort = TRUE) |>
  dplyr::rename(
    entity = AU,
    n = n
  )
summary_tbl <- add_h_index(summary_tbl)
summary_tbl <- add_aac(summary_tbl)
```

2.5. AAC used to determine dominant strength over the next two elements

The AAC was used to quantify the dominance strength of the leading entity relative to the next two ranked entities. Let v_1, v_2, and v_3 denote the observed values of the first-, second-, and third-ranked entities, respectively.

The dominance ratio was calculated as $r = (v_1 / v_2)/(v_2 / v_3)$, and the AAC was then computed as $AAC = r/(1 + r)$. Thus, AAC transforms the dominance ratio into a bounded coefficient between 0 and 1.

A higher AAC indicates stronger dominance of the leading entity over its nearest competitors, whereas a lower AAC suggests a more even distribution among the top-ranked entities. App4WoSCCinR provides an AAC dashboard to support visual interpretation of dominance patterns across bibliometric domains (Fig. 1) [33].

2.6. Multi-panel output generation

Domain-specific tables are routed to plotting functions and then assembled into exportable outputs.

```
p1 <- render_summary_plot(country_tbl)
p2 <- render_slope_plot(year_tbl)
p3 <- render_chord_plot(collab_tbl)
p4 <- render_ssplot(keyword_tbl)
export_plots(list(p1, p2, p3, p4), outdir = tempdir())
```

These examples are intended to illustrate the internal logic of the framework rather than to reproduce the full application code.

2.7. Repository structure

The repository is organized to reflect the modular architecture of the software. Core scripts and functions are grouped into components for input handling, metadata preprocessing, analytical summarization, AAC computation, visualization generation, and export. This structure helps readers identify the major implementation units responsible for parsing WoSCC data, constructing domain summaries, rendering Figures, and packaging outputs.

2.8. User-facing workflow and functionalities

At the interface level, App4WoSCCinR provides a simplified two-step workflow(Fig.1):

1. **Upload WoSCC data**
2. Users may either upload a WoSCC file (.xls/.xlsx) or provide a valid URL to bibliometric data, using the passcode 5730687835; a demo dataset is also available for rapid testing without a passcode.
3. **Run analysis**
4. By clicking the analysis function, the system automatically performs preprocessing, summary computation, and visualization rendering in real time.

The main output functions include:

- **Report**, which generates a structured summary of bibliometric performance;
- **Slope**, which displays temporal publication trends;
- **Chord**, which visualizes inter-entity relationships;

Table 2
Internal-validation of directly comparable outputs from Fig. 3 against WoSCC for the same SoftwareX dataset.

Output/domain	Definition used for validation	App4WoSCCinR result	WoSCC result	Agreement	Note
Total records	Number of retrieved SoftwareX records (2017–2026)	2,061	2,061	Exact	Same WoSCC search set
Overall h-index	h-index for the same retrieved record set	54	54	Exact	Same citation-report basis
Top country	Country with the highest publication count	United States = 304	United States = 304	Exact	Based on WoSCC address metadata
Top journal	Journal with the highest publication count	SoftwareX = 2,061	SoftwareX = 2,061	Exact	Dataset restricted to SoftwareX
Top publication year	Year with the highest publication count	2025 = 474	2025 = 474	Exact	Same year field in WoSCC
Top document type	Document type with the highest publication count	Article = 2,020	Article = 2,020	Exact	Same WoSCC document-type label

Note. Other domain comparison in Supplemental Appendix 2

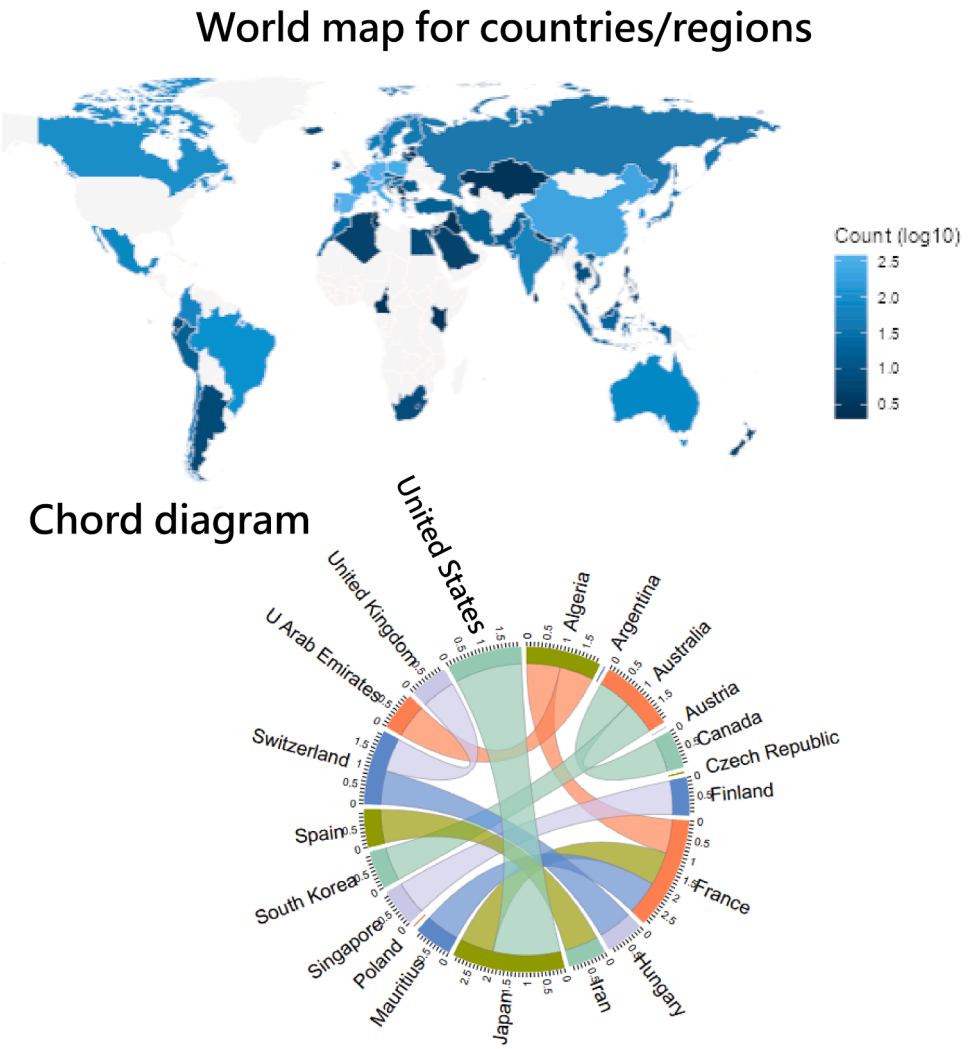


Fig. 3. Global distribution and collaboration visualization(note. the upper panel shows country-level publication counts on a world map, while the lower chord diagram shows international collaboration patterns among major contributing countries).

- **SSplot**, which presents cluster structure and dominance;
- **Map / China / USA**, which provide geographic publication distributions; and
- **Lotka**, which models the distribution of author productivity.

Optional download functions allow users to export Figures as PNG files and to save all outputs as a ZIP archive. Depending on dataset size and plotting complexity, execution time typically ranges from several seconds for small demonstration files to a few minutes for larger WoSCC datasets.

2.9. Sample code snippets

2.9.1. R code for visualization and reporting

App4WoSCCinR was implemented on shiny.io[34] to support automated bibliometric visualization and reporting from standard 72-column WoSCC metadata.

The framework integrates data preprocessing, metadata summarization, and visual output generation within a unified workflow. Rather than relying solely on single-purpose scripts for individual plots, this study used modular R code to produce a summary report and multiple visualization outputs across key metadata domains, including country, institution, department, author, journal, year, document type, discipline, keyword, and region.

In this study, the generated outputs included summary plots, temporal trend plots, chord diagrams, Sankey diagrams, SSplots, geographic maps, and other bibliometric visualizations.

Following the reporting structure suggested in previous bibliometric studies [8,30], the results were organized into two parts: (1) performance analysis findings and (2) visualization outputs.

2.9.2. R code used in this study

All data processing, visualization, and statistical analyses were conducted using R version 4.3.3 [35]. The application was developed and tested in the RStudio integrated development environment [36]. Core functions were written to automate metadata cleaning, restructuring, summary reporting, dominance analysis, and Figure generation, thereby allowing users to complete bibliometric analysis through a simple upload-and-run web interface.

The code base supports reproducible generation of publication indicators, AAC-based rankings, and over 100 visualizations from WoSCC-exported metadata, as deposited in Github [37]. Instruction with MP4 video[38] and An example of url link [39] to App4WoSCCinR are presented.

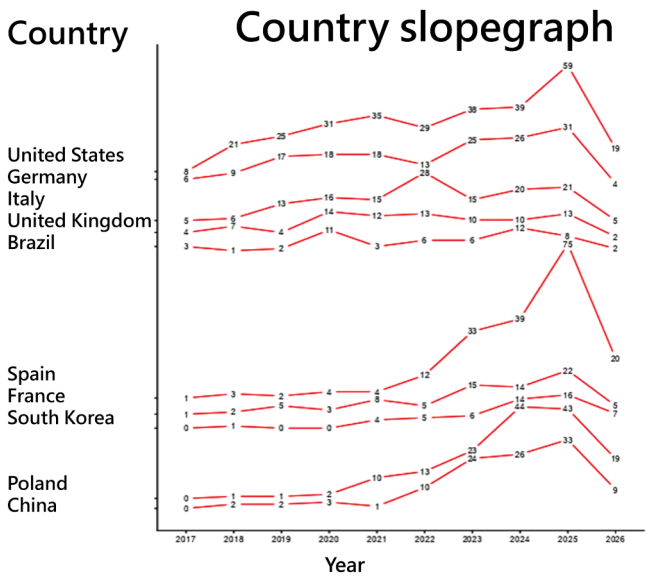


Fig. 4. Temporal trends of leading countries in SoftwareX publications from 2017 to 2026(Note. countries are vertically offset to avoid overlap among longitudinal trend lines and countries are ordered according to overall publication contribution, with higher-output countries placed first).

3. Illustrative examples

3.1. Data source

Articles published in SoftwareX were retrieved from WoSCC, including both the Science Citation Index Expanded (SCI-Expanded) and the Social Sciences Citation Index (SSCI). The coverage period was from 2017 to 2026, with the search conducted on March 20, 2026. By specifying both the subsets and the temporal coverage, this study ensures full transparency and facilitates reproducibility for future bibliometric investigations. Study data are deposited in Github [31] and Supplemental Appendix 1, with buttons of Browse and Run WoS to show visualizations in Fig. 1 .

3.2. Demonstrations of two results via App4WoSCCin R

3.2.1. Performance analysis findings

Table 1 table lists the leading entities across 10 WoSCC metadata domains, including h-index, publication count, and AAC. AAC indicates the dominance of the leading entity relative to the next-ranked competitors within each domain, including 2,061 articles with an overall h-index of 54 and incorporates publication counts (n), h-index values (h) [33], and AAC [8,27,28,34] scores to quantify dominance.

The results indicate that the United States is the leading country (n = 304, h = 30, AAC = 0.58). Delft University of Technology (Netherlands) is the most prolific institution (n = 16, h = 6, AAC = 0.53), while the Department of Computer Science is the most prevalent affiliation (n = 89, h = 11, AAC = 0.77), and SoftwareX is the dominant journal, as

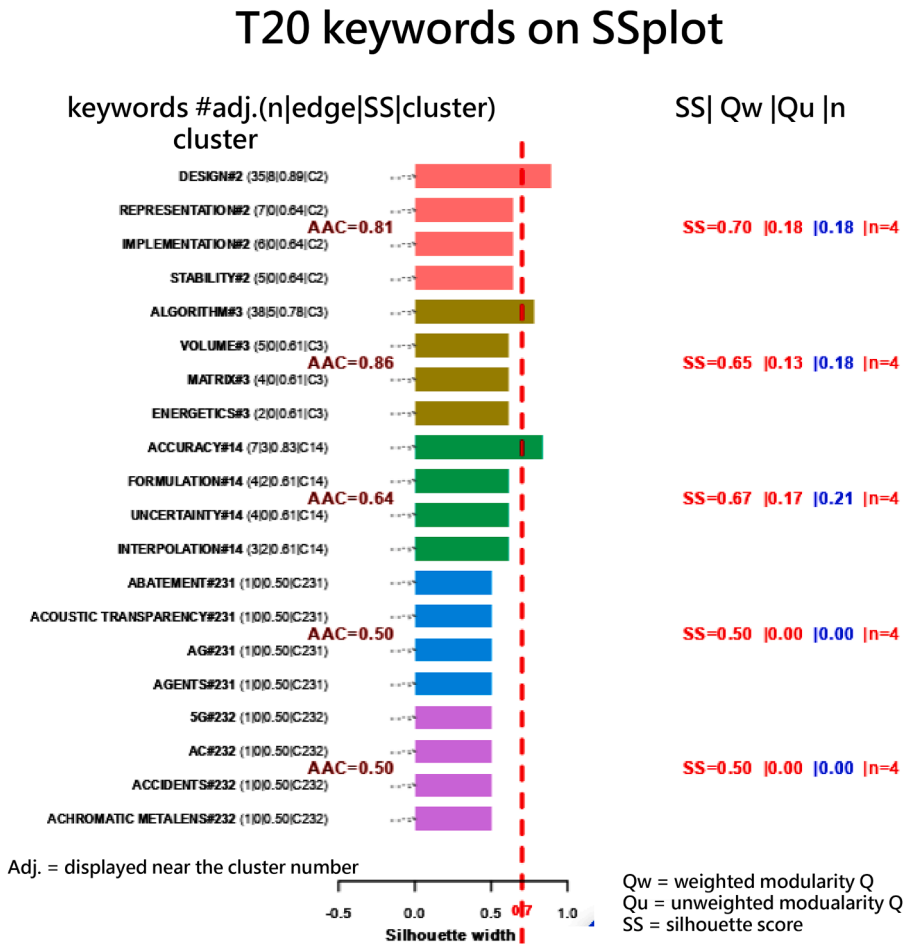


Fig. 5. SSplot visualization of FLCA-SIL-MA clustering results[25] for top keywords in SoftwareX publications(Note. the SSplot enables rapid visual interpretation of keyword clusters, AAC values, silhouette scores, and cluster structures at a glance).

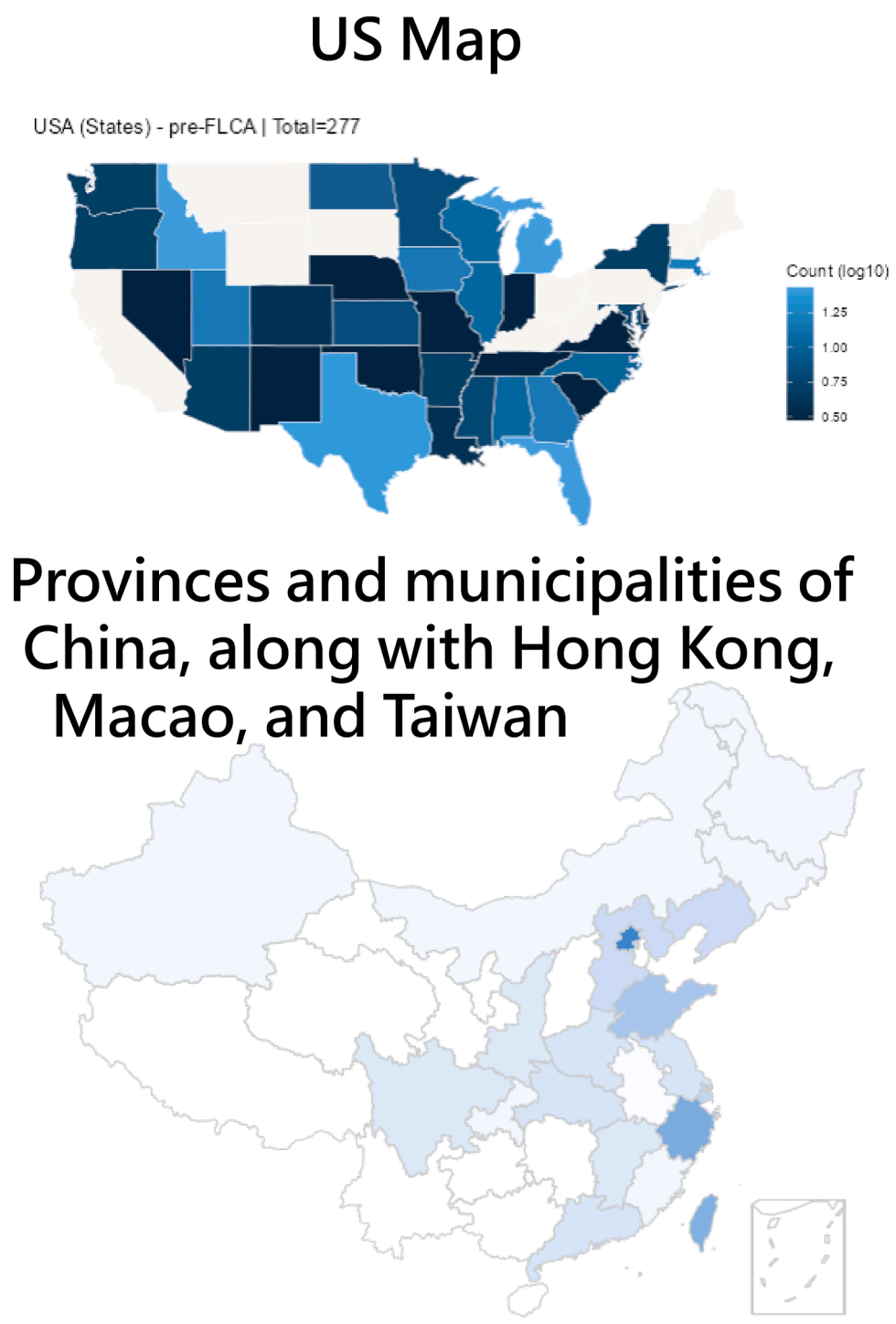


Fig. 6. Geographic distribution at regional level(note. the two maps show regional publication distributions across U.S. states and Chinese provinces/municipalities, with color intensity representing publication counts).

expected from the dataset. Others are shown in Table 1.

Table 2 presents outputs that are directly comparable to native WoSCC summary fields and were therefore expected to match exactly when the same record set and counting rules were used. Other metadata comparisons are presented in Supplementary Table S1.

3.2.2. *Visualization outputs*

App4WoSCCinR generates a world map in Fig. 3 showing publication counts by country, alongside a chord diagram illustrating international collaboration patterns among major contributing countries.

The Fig. 4 displays yearly publication counts for the top contributing countries identified from WoSCC metadata. Red lines connect annual publication counts to illustrate longitudinal growth patterns. The United States demonstrated the highest and most sustained publication activity, whereas countries such as Spain, Poland, and China exhibited rapid increases in recent years. The visualization highlights dynamic shifts in international research productivity over time.

Keywords were clustered using the FLCA. Colored bars represent cluster membership, whereas AAC values indicate dominance of the leading keyword within each cluster. Silhouette scores (SS) and

Table 3
Feature-by-feature comparison of App4WoSCCinR with widely used bibliometric tool.

Feature / criterion	App4WoSCCinR	CiteSpace	VOSviewer	bibliometrix / biblioshiny	BibexPy	BiblioMerge
Primary role	WoSCC upload-to-report	Mapping, visualization, exploration	Network, visualization	Bibliometric analysis in R	Preprocessing, merging, deduplication data	Merging, deduplication, normalizing data
Handling Excel (.xls/.xlsx)	**Yes**	No	No	Partial	No	No
No-code workflow	**Yes**	Partial	Partial	**Yes** (biblioshiny)	No	No
Preprocessing within workflow	**Yes**	Partial	Limited	Partial	**Yes**	**Yes**
Reduce manual wrangling data	**Yes**	No	No	Partial	Partial	Partial
Multi-domain summary report	**Yes**	Partial	No	Partial	No	No
Diverse visual outputs in one run	**Yes**	Partial	Partial	Partial	No	No
Network-mapping visualizations	**Yes**	**Yes**	**Yes**	**Yes**	No	No
Geographic, temporal outputs	**Yes**	Partial	Limited	Partial	No	No
Merge and deduplicate records	Partial	No	No	Partial	**Yes**	**Yes**
AAC-based dominance reporting	**Yes**	No	No	No	No	No
Exportable report / Figure package	**Yes**	Partial	Partial	Partial	No	No
URL-Link repot	**Yes**	No	No	No	No	No
2-Step in a run	**Yes**	No	No	No	No	No

Note. Yes = explicitly supported as a core workflow element; Partial = available in a narrower, indirect, or less integrated form; No = not a primary or explicitly documented function in the compared source. Replace "Table X" with the final table number used in the manuscript.

modularity-related indicators are shown on the right to evaluate cluster cohesion and separation. Higher AAC and SS values indicate stronger thematic dominance and more stable cluster structures.

Fig. 5.

The maps in Fig. 6 display publication distributions across U.S. states and Chinese provinces/municipalities (including Hong Kong, Macao, and Taiwan), providing finer-grained spatial insights into research activity.

Based on Table 3, the author productivity distribution only partly follows Lotka's law.

3.3. Reaching the study goals

This study achieves the study goals that a streamlined web-based framework, implemented through App4WoSCCinR, can effectively (1) perform bibliometric analysis from standard 72-column WoSCC Excel metadata and (2) generate outputs with substantial analytical value.

As shown in Table 3, the practical contribution of App4WoSCCinR is not to replace established science-mapping tools, but to provide an integrated, low-barrier workflow that accepts standard WoSCC Excel exports directly and generates structured summary reporting plus diverse visual outputs within a single web-based execution environment.

Through its two-step workflow—data upload and automated analysis—the system produced both a structured performance summary and over 100 visualizations across multiple metadata domains.

These findings demonstrate that App4WoSCCinR is capable of supporting bibliometric interpretation in a manner that is clear, efficient, and reproducible. In particular, the integration of the AAC [8,20,21,24,27] enabled more nuanced identification of dominant entities than simple frequency rankings alone.

The proposed framework thus meets the study goal of providing a practical and transparent alternative for bibliometric analysis, especially for users seeking low-barrier and reproducible analytical tools.

4. Impact

App4WoSCCinR extends the earlier 10StepsinExcelforWoS framework [8] into a visualization-oriented bibliometric platform capable of processing standard 72-column WoSCC spreadsheet metadata through a browser-based workflow. By integrating metadata ingestion, preprocessing, structured reporting, AAC-enhanced dominance analysis, and multi-domain visualization within a single environment, the platform reduces manual preprocessing requirements and lowers technical barriers for bibliometric exploration.

Compared with established tools such as CiteSpace, VOSviewer, and Bibliometrix, the contribution of App4WoSCCinR lies primarily in workflow integration and direct handling of native WoSCC spreadsheet metadata rather than replacement of existing science-mapping software. The framework emphasizes rapid upload-and-run processing together with exportable structured summaries and diverse visualization outputs across countries, institutions, authors, journals, keywords, and temporal trends.

The inclusion of AAC [8,26,27,36] provides a complementary dominance-oriented indicator that extends conventional frequency-based summaries by quantifying the relative strength of leading entities compared with nearby competitors.

App4WoSCCinR also improves transparency of bibliometric preprocessing by documenting metadata normalization and structured aggregation workflows. This is particularly relevant because WoSCC metadata may contain inconsistencies such as incomplete affiliations, author-name ambiguity, inconsistent document-type labeling, and incomplete keyword coverage in historical records [40]. Although these issues cannot be fully eliminated by downstream software, deterministic preprocessing and normalization procedures can improve interpretability and reproducibility.

Several limitations should be addressed. First, the current implementation primarily supports WoSCC-derived metadata and does not yet provide direct interoperability with Scopus, Dimensions, or OpenAlex formats. Second, preprocessing outcomes may still be influenced by metadata incompleteness, author-disambiguation ambiguity, and database-specific indexing practices. Third, formal scalability benchmarking for very large bibliometric datasets has not yet been systematically evaluated. These limitations remain important directions for future development and validation.

5. Conclusion

App4WoSCCinR offers a practical web-based framework for analyzing standard 72-column WoSCC metadata through a simple two-step workflow. By generating structured reports, diverse visualizations, and AAC-based dominance indicators without programming, it improves accessibility, transparency, and reproducibility for efficient bibliometric analysis.

Data availability

The data are public and available at Github.

CRediT authorship contribution statement

Tsair-Wei Chien: Methodology. **Lifan Chen:** Data curation, Conceptualization. **Willy Chou:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We would like to acknowledge Github which allowing us to deposit our R code and datasets at no charges.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.softx.2026.102770.

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